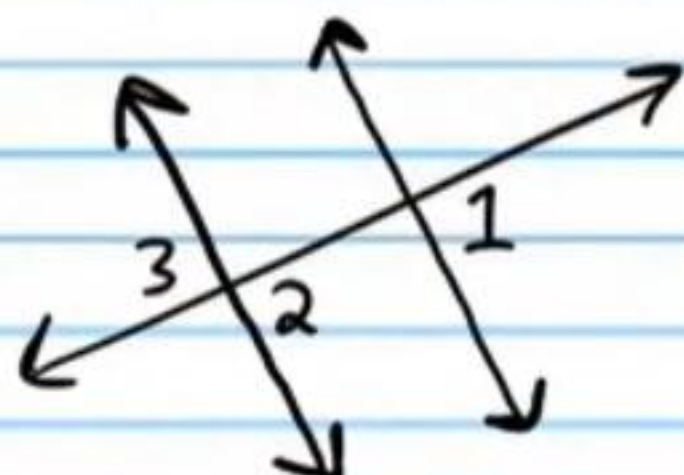


Proofs and Properties of Equality and Congruence (Part II).

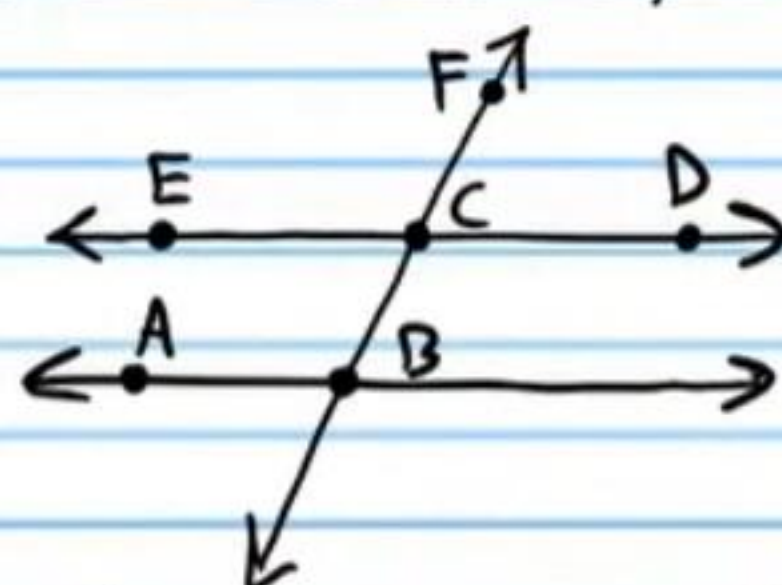
Prove the following statements with two-column proofs.

① If $m\angle 1 = m\angle 2$ in the figure, then $m\angle 1 = m\angle 3$.



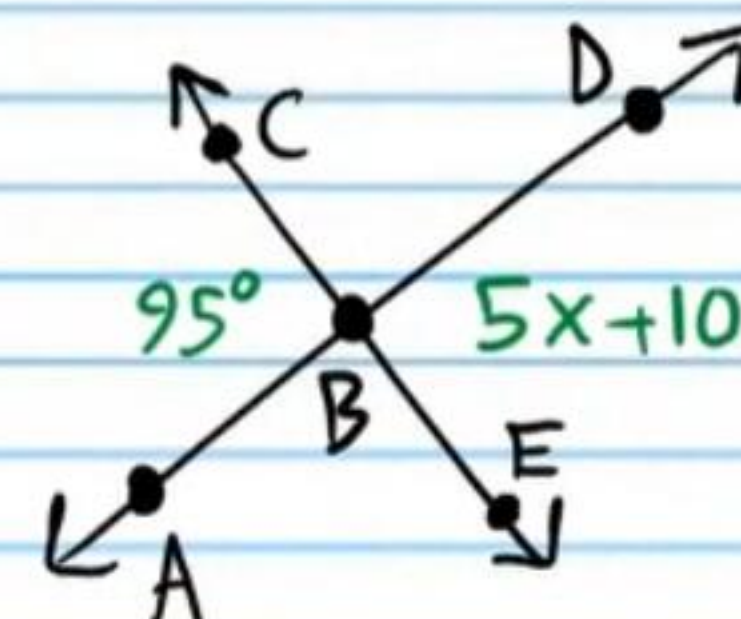
Statement	Reason
I) $m\angle 1 = m\angle 2$	Given.
II) $m\angle 2 = m\angle 3$	Vert. $\angle$'s Thm.
III) $m\angle 1 = m\angle 3$	Trans. Prop of =.

② If $m\angle ABC = m\angle BCD$, then $m\angle ABC = m\angle ECF$.



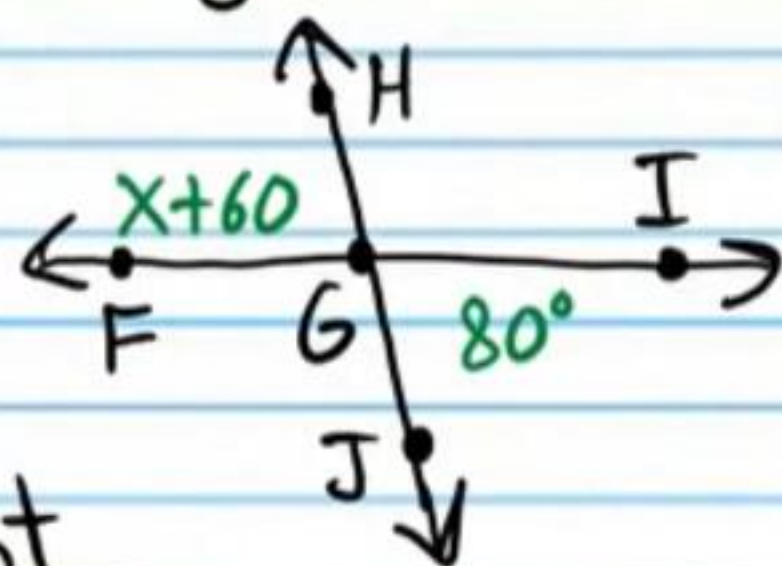
Statement	Reason
I) $m\angle ABC = m\angle BCD$	Given.
II) $m\angle BCD = m\angle ECF$	Vert. $\angle$'s Thm.
III) $m\angle ABC = m\angle ECF$	Trans. Prop of =.

③ If $\angle ABC$ and $\angle EBD$ are vertical angles in the figure below, then $x = 17$.



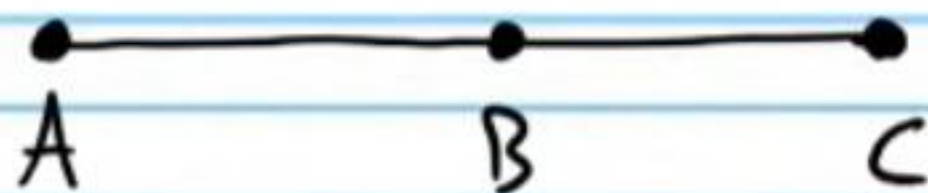
Statement	Reason
I) $\angle ABC$ & $\angle EBD$ are vert. $\angle$'s	Given.
II) $m\angle ABC = m\angle EBD$	Vert. $\angle$'s Thm.
III) $m\angle ABC = 95^\circ$ and $m\angle EBD = 5x + 10$	Given (diagram).
IV) $95 = 5x + 10$	Subs. Prop of =.
V) $85 = 5x$	Subst. Prop of =.
VI) $17 = x$	Div Prop of =.
VII) $x = 17$	Symm. Prop of =.

④ If $\angle FGH$ and $\angle JGI$ in the figure below are vertical angles, then $x = 20$.



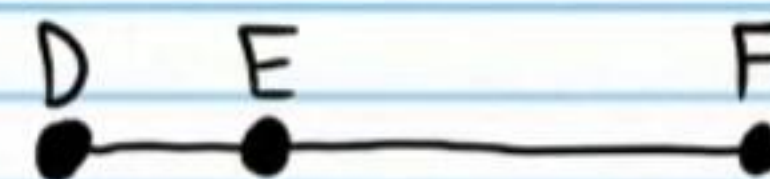
Statement	Reason
I) $\angle FGH$ & $\angle JGI$ are vert $\angle$'s.	Given.
II) $m\angle FGH = m\angle JGI$	Vert. $\angle$'s Thm.
III) $m\angle FGH = x+60$ and $m\angle JGI = 80$	Given (diagram).
IV) $x+60 = 80$	Subs. Prop of =.
V) $x = 20$	Subt. Prop of =.

⑤ If $AB = 9$ and $BC = 7$ in the segment below, then $AC = 16$.



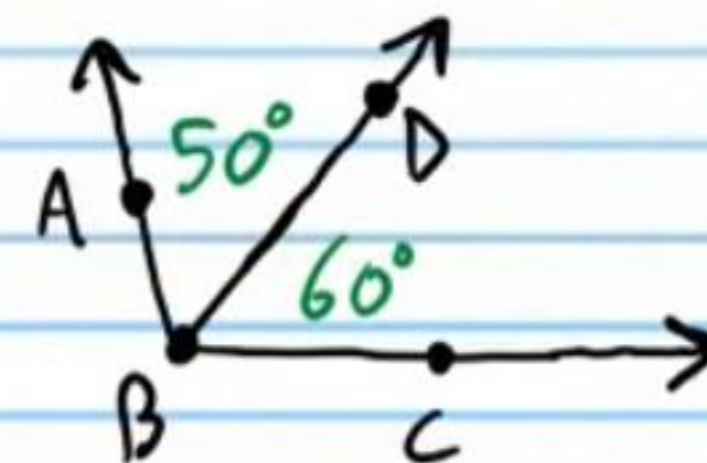
Statement	Reason
I) $AB = 9$ & $BC = 7$	Given.
II) $AC = AB + BC$	Seg Add Post.
III) $AC = 9 + 7$	Subs. Prop of =.
IV) $AC = 16$	Simplification.

⑥ If $DE = 3$ and $EF = 8$, then $DF = 11$.



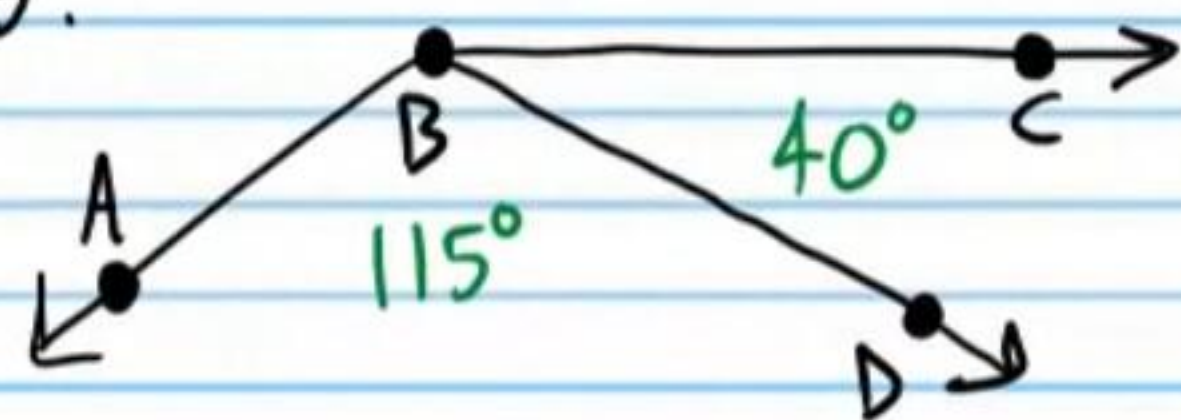
Statement	Reason
I) $DE = 3$ & $EF = 8$	Given.
II) $DF = DE + EF$	Seg Add Post.
III) $DF = 3 + 8$	Subs. Prop of =.
IV) $DF = 11$	Simplification.

⑦ If $m\angle ABD = 50^\circ$ and $m\angle DBC = 60^\circ$, then $m\angle ABC = 110^\circ$.



Statement	Reason
I) $m\angle ABD = 50^\circ$ & $m\angle DBC = 60^\circ$	Given.
II) $m\angle ABC = m\angle ABD + m\angle DBC$	$\angle$ Add Post.
III) $m\angle ABC = 50 + 60$	Subs. Prop of =.
IV) $m\angle ABC = 110^\circ$	Simplification.

⑧ If $m\angle ABD = 115^\circ$ and $m\angle DBC = 40^\circ$, then $m\angle ABC = 155^\circ$.



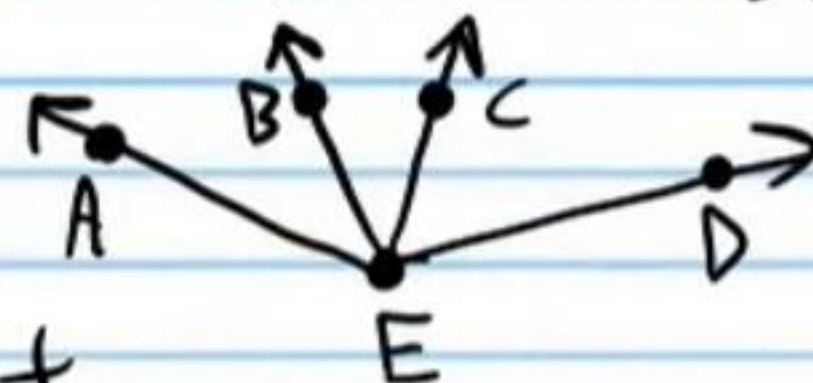
Statement	Reason
I) $m\angle ABD = 115^\circ$ & $m\angle DBC = 40^\circ$	Given.
II) $m\angle ABC = m\angle ABD + m\angle DBC$	$\angle$ Add Post.
III) $m\angle ABC = 115 + 40$	Subs. Prop of =.
IV) $m\angle ABC = 155^\circ$	Simplification.

⑨ If B and C lie between A and D (collinear) on the segment below, then $AD = AB + BC + CD$.



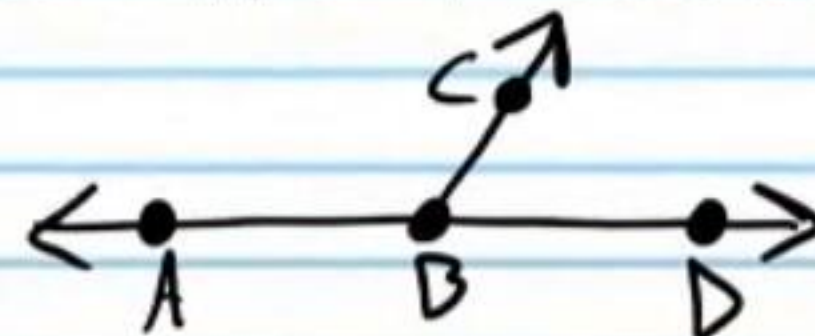
Statement	Reason
I) B & C lie between A & D	Given.
II) $AD = AC + CD$	Seg Add Post.
III) $AC = AB + BC$	Seg Add Post.
IV) $AD = AB + BC + CD$	Subs. Prop of =.

⑩ $m\angle AED = m\angle AEB + m\angle BEC + m\angle CED$.



Statement	Reason
I) Diagram	Given.
II) $m\angle AED = m\angle AEC + m\angle CED$	$\angle$ Add Post.
III) $m\angle AEC = m\angle AEB + m\angle BEC$	$\angle$ Add Post.
IV) $m\angle AED = m\angle AEB + m\angle BEC + m\angle CED$	Subs. Prop of =.

⑪ If $m\angle ABC = 130^\circ$ in the diagram below, then $m\angle CBD = 50^\circ$.



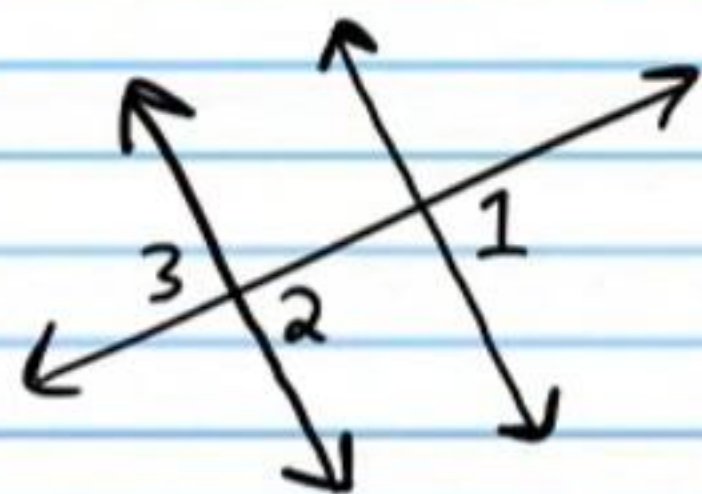
Statement	Reason
I) $\angle ABC$ & $\angle CBD$ form a linear pair.	Given (diagram).
II) $\angle ABC$ & $\angle CBD$ are supplementary.	Linear pair Post.
III) $m\angle ABC + m\angle CBD = 180$	Def. of supp $\angle$'s.
IV) $m\angle ABC = 130^\circ$	Given.
V) $130 + m\angle CBD = 180$	Subs. Prop of =.
VI) $m\angle CBD = 50^\circ$	Subst. Prop of =.

⑫ If $m\angle GEF = 60^\circ$ in the diagram below, then $m\angle DEG = 120^\circ$.



Statement	Reason
I) $\angle GEF$ & $\angle DEG$ form a linear pair	Given (diagram).
II) $\angle GEF$ & $\angle DEG$ are supplementary.	Linear Pair Post.
III) $m\angle GEF + m\angle DEG = 180$	Def. of supp $\angle$'s.
IV) $m\angle GEF = 60$	Given.
V) $60 + m\angle DEG = 180$	Subst. Prop of =.
VI) $m\angle DEG = 120^\circ$	Subst. Prop of =.

⑬ The converse of problem #1. That is, if $m\angle 1 = m\angle 3$, then $m\angle 1 = m\angle 2$.



Statement	Reason
I) $m\angle 1 = m\angle 3$	Given.
II) $m\angle 3 = m\angle 2$	Vert. $\angle$'s Thm
III) $m\angle 1 = m\angle 2$	Trans. Prop of =.

⑭ The Transitive Property of Congruence: If $\overline{AB} \cong \overline{CD}$, and $\overline{CD} \cong \overline{EF}$, then $\overline{AB} \cong \overline{EF}$.

Statement	Reason
I) $\overline{AB} \cong \overline{CD}$ & $\overline{CD} \cong \overline{EF}$	Given.
II) $AB = CD$ & $CD = EF$	Def. of $\cong$ Seg
III) $AB = EF$	Trans. Prop of =.
IV) $\overline{AB} \cong \overline{EF}$	Def. of $\cong$ Seg

⑮ If $\angle A \cong \angle B$ and $\angle B \cong \angle C$, then $\angle A \cong \angle C$.

Statement	Reason
I) $\angle A \cong \angle B$ & $\angle B \cong \angle C$	Given.
II) $m\angle A = m\angle B$ & $m\angle B = m\angle C$	Def. of $\cong \angle$'s.
III) $m\angle A = m\angle C$	Trans. Prop of =.
IV) $\angle A \cong \angle C$	Def. of $\cong \angle$'s.

⑩ If $\vec{BD}$ bisects $\angle ABC$, then $m\angle DBC = \frac{1}{2}m\angle ABC$.



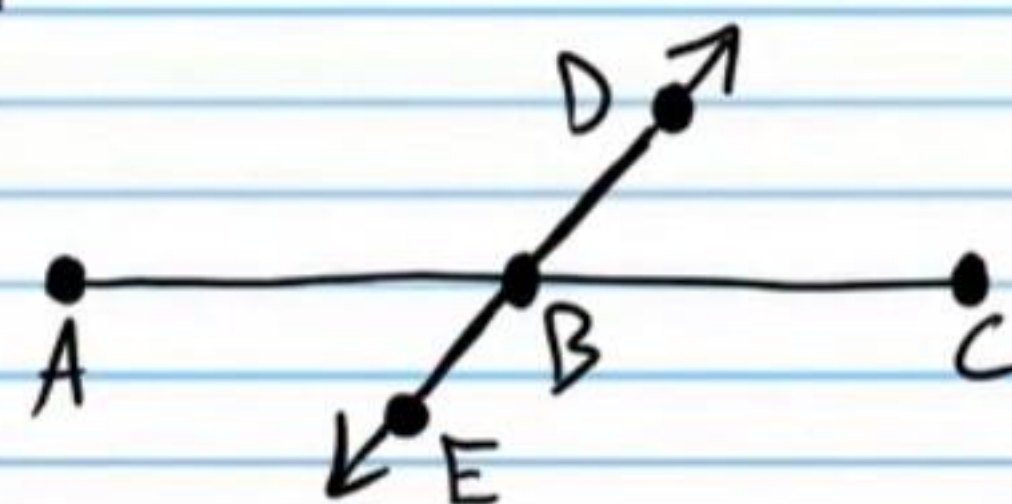
Statement	Reason
I) $\vec{BD}$ bisects $\angle ABC$.	Given.
II) $m\angle ABD = m\angle DBC$	Def. of $\angle$ bisector.
III) $m\angle ABC = m\angle ABD + m\angle DBC$	$\angle$ Add Post.
IV) $m\angle ABC = m\angle DBC + m\angle DBC$	Subs. Prop of =.
V) $m\angle ABC = 2m\angle DBC$	Simplification.
VI) $\frac{m\angle ABC}{2} = m\angle DBC$	Div. Prop of =.
VII) $m\angle DBC = \frac{1}{2}m\angle ABC$	Symm Prop of =.

⑪ If M is the midpoint of the segment below, then $AB = 2MB$.



Statement	Reason
I) M is the midpoint of $\overline{AB}$	Given.
II) $AM = MB$	Def. of midpoint.
III) $AB = AM + MB$	Seg Add Post.
IV) $AB = MB + MB$	Subs. Prop of =.
V) $AB = 2MB$	Simplification.

⑮ If $\overleftrightarrow{DE}$ bisects $\overline{AC}$ in the figure below, then $\overline{AB} \cong \overline{BC}$.



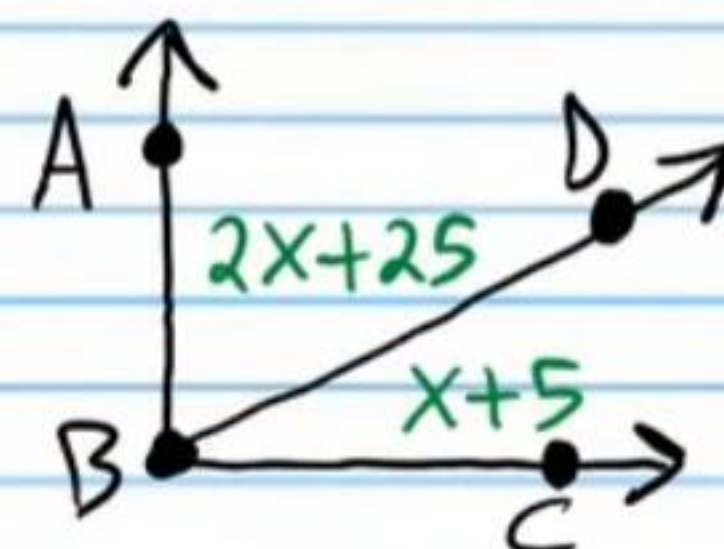
Statement	Reason
I) $\overleftrightarrow{DE}$ bisects $\overline{AC}$.	Given.
II) $AB = BC$	Def. of seg bisector.
III) $\overline{AB} \cong \overline{BC}$	Def. of $\cong$ seg

⑯ If $\overline{AB} \cong \overline{CE}$ in the segments below, then $AB = CD + DE$.



Statement	Reason
I) $\overline{AB} \cong \overline{CE}$	Given.
II) $AB = CE$	Def. of $\cong$ seg
III) $CE = CD + DE$	Seg Add Post.
IV) $AB = CD + DE$	Trans. Prop of =.

⑰ If $\angle ABC$ below is a right angle, then $m\angle ABD = 65^\circ$.



Statement	Reason
I) $\angle ABC$ is a rt. $\angle$.	Given.
II) $m\angle ABC = 90^\circ$	Def. of a rt $\angle$.
III) $m\angle ABC = m\angle ABD + m\angle DBC$	$\angle$ Add Post.
IV) $m\angle ABD = 2x + 25$ and $m\angle DBC = x + 5$	Given (diagram).
V) $90 = 2x + 25 + x + 5$	Subst. Prop of =.
VI) $90 = 3x + 30$	Simplification.
VII) $60 = 3x$	Subst. Prop of =.
VIII) $20 = x$	Div Prop of =.
IX) $m\angle ABD = 2(20) + 25$	Subst. Prop of =.
X) $m\angle ABD = 65^\circ$	Simplification.